Answer following question

1 :- Describe all three data set in terms of following : how many features, their categories (num,text,categorical, nominal,ordinal etc), number of tuples, missing values.

Ans :- To describe three datasets, we analyze:

Number of features Feature categories (Numerical, Categorical, Nominal, Ordinal, Text) Number of tuples Missing values

```
import pandas as pd

# ---------------- Dataset 1 ----------------
data1 = {
    "Age": [20, 21, 22, None],
    "Gender": ["Male", "Female", "Male", "Female"],
    "Salary": [20000, 25000, None, 30000],
    "City": ["Mumbai", "Pune", "Delhi", "Mumbai"],
    "Feedback": ["Good", "Excellent", "Average", "Good"]
}
df1 = pd.DataFrame(data1)

# ---------------- Dataset 2 ----------------
data2 = {
    "Marks": [80, 90, 85, None],
    "Grade": ["A", "A+", "B", "A"],
    "Subject": ["Math", "Science", "English", "Math"],
    "Remarks": ["Good", "Very Good", "Average", "Excellent"]
}
df2 = pd.DataFrame(data2)

# ---------------- Dataset 3 ----------------
data3 = {
    "Product_ID": ["P1", "P2", "P3", "P4"],
    "Price": [100, 200, 150, None],
    "Quantity": [2, 3, 1, 5],
    "Category": ["Electronics", "Clothing", "Electronics", "Clothing"],
    "Rating": [5, 4, None, 3],
    "Review": ["Good", "Nice", "Average", "Excellent"]
}
df3 = pd.DataFrame(data3)


# Function to analyze dataset
def analyze(df, name):
    print("\nDataset:", name)
    print("Features:", df.shape[1])
    print("Tuples:", df.shape[0])

    print("\nFeature Types:")
    for col in df.columns:
        if df[col].dtype in ['int64', 'float64']:
            print(col, "-> Numerical")
        else:
            print(col, "-> Categorical/Text")

    print("\nMissing Values:")
    print(df.isnull().sum())


# Analyze all datasets
analyze(df1, "Dataset 1")
analyze(df2, "Dataset 2")
analyze(df3, "Dataset 3")
```

```
Dataset: Dataset 1
Features: 5
Tuples: 4

Feature Types:
Age -> Numerical
Gender -> Categorical/Text
Salary -> Numerical
```

```
City -> Categorical/Text
Feedback -> Categorical/Text

Missing Values:
Age         1
Gender      0
Salary      1
City        0
Feedback    0
dtype: int64

Dataset: Dataset 2
Features: 4
Tuples: 4

Feature Types:
Marks -> Numerical
Grade -> Categorical/Text
Subject -> Categorical/Text
Remarks -> Categorical/Text

Missing Values:
Marks      1
Grade      0
Subject    0
Remarks    0
dtype: int64

Dataset: Dataset 3
Features: 6
Tuples: 4

Feature Types:
Product_ID -> Categorical/Text
Price -> Numerical
Quantity -> Numerical
Category -> Categorical/Text
Rating -> Numerical
Review -> Categorical/Text

Missing Values:
Product_ID    0
Price         1
Quantity      0
Category      0
Rating        1
Review        0
dtype: int64
```

2 :- In IRIS data set which non graphical EDA is performed ?

Ans :- The Iris dataset is a famous dataset containing:

150 samples (tuples)

4 numerical features:

Sepal Length

Sepal Width

Petal Length

Petal Width

1 categorical feature:

Species (Setosa, Versicolor, Virginica)

Types of Non-Graphical EDA

1. Summary Statistics

Mean

Median

Standard Deviation

Min & Max

example of code

```
import pandas as pd
from sklearn.datasets import load_iris

iris = load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)

print(df.describe())
```

```
       sepal length (cm)  sepal width (cm)  petal length (cm)  \
count         150.000000        150.000000         150.000000
mean            5.843333          3.057333           3.758000
std             0.828066          0.435866           1.765298
min             4.300000          2.000000           1.000000
25%             5.100000          2.800000           1.600000
50%             5.800000          3.000000           4.350000
75%             6.400000          3.300000           5.100000
max             7.900000          4.400000           6.900000

       petal width (cm)
count        150.000000
mean           1.199333
std            0.762238
min            0.100000
25%            0.300000
50%            1.300000
75%            1.800000
max            2.500000
```

. Checking Missing Values

```
print(df.isnull().sum())
```

```
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
dtype: int64
```

3. Data Types

```
print(df.dtypes)
```

```
sepal length (cm)    float64
sepal width (cm)     float64
petal length (cm)    float64
petal width (cm)     float64
dtype: object
```

4. Unique Values (Categorical Analysis)

```
df['species'] = iris.target
print(df['species'].value_counts())
```

```
species
0    50
1    50
2    50
Name: count, dtype: int64
```

5. Correlation Analysis

```
print(df.corr())
```

```
                   sepal length (cm)  sepal width (cm)  petal length (cm)  \
sepal length (cm)           1.000000         -0.117570           0.871754
sepal width (cm)           -0.117570          1.000000          -0.428440
petal length (cm)           0.871754         -0.428440           1.000000
petal width (cm)            0.817941         -0.366126           0.962865
species                     0.782561         -0.426658           0.949035

                   petal width (cm)   species
sepal length (cm)          0.817941  0.782561
sepal width (cm)          -0.366126 -0.426658
petal length (cm)          0.962865  0.949035
petal width (cm)           1.000000  0.956547
species                    0.956547  1.000000
```

6. Skewness & Distribution

```
print(df.skew())
```

```
sepal length (cm)    0.314911
sepal width (cm)     0.318966
petal length (cm)   -0.274884
petal width (cm)    -0.102967
species              0.000000
dtype: float64
```

3 :- In Air Quality data set which graph is used to perform multivariate analysis?

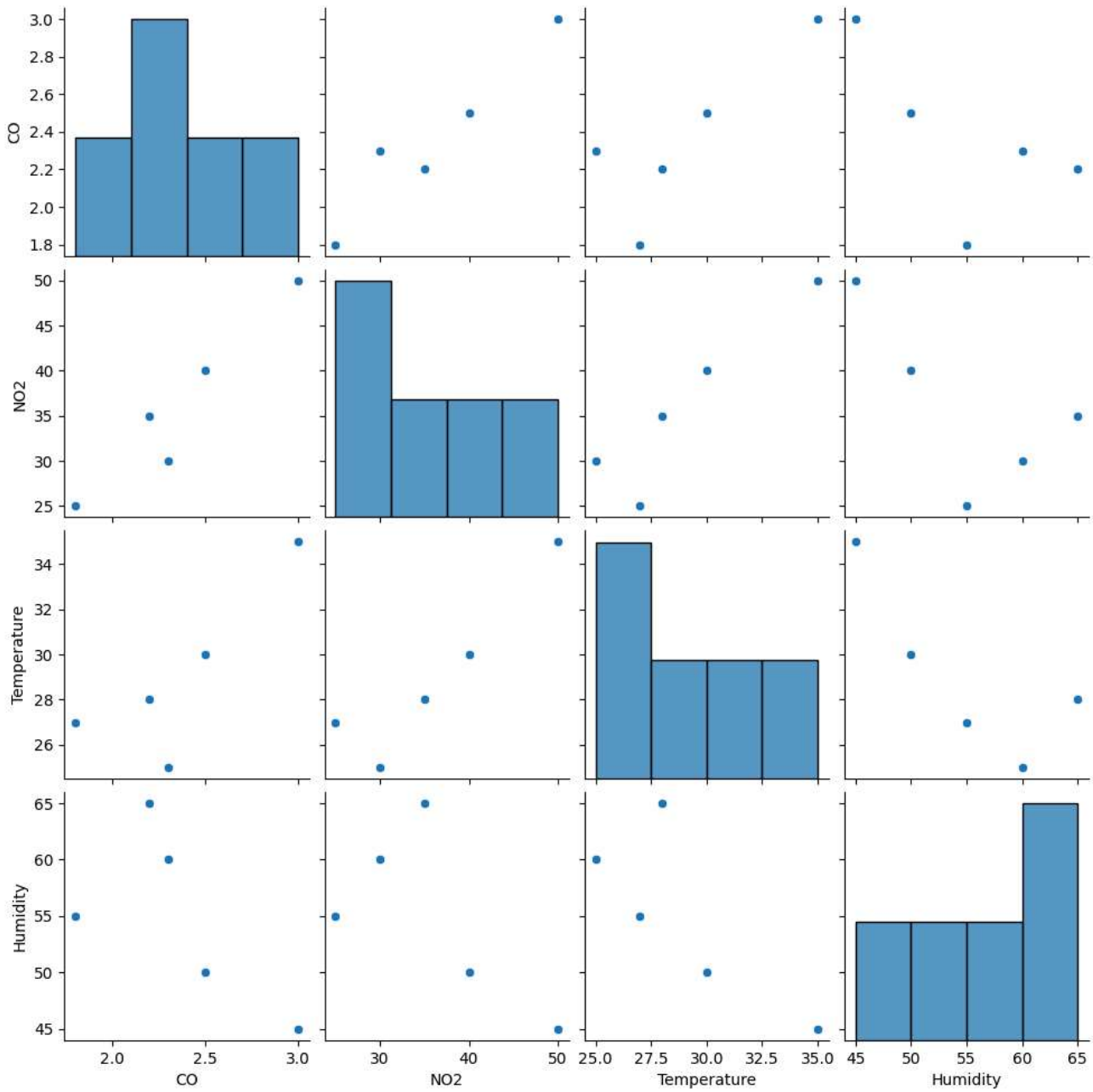
Ans :- The Air Quality dataset contains multiple variables such as pollutants (CO, NO_2, SO_2), temperature, humidity, and wind speed. To analyze the relationship between more than two variables simultaneously, multivariate analysis is performed.

1. Pair Plot (Scatter Plot Matrix)

The most commonly used graph for multivariate analysis is the Pair Plot.

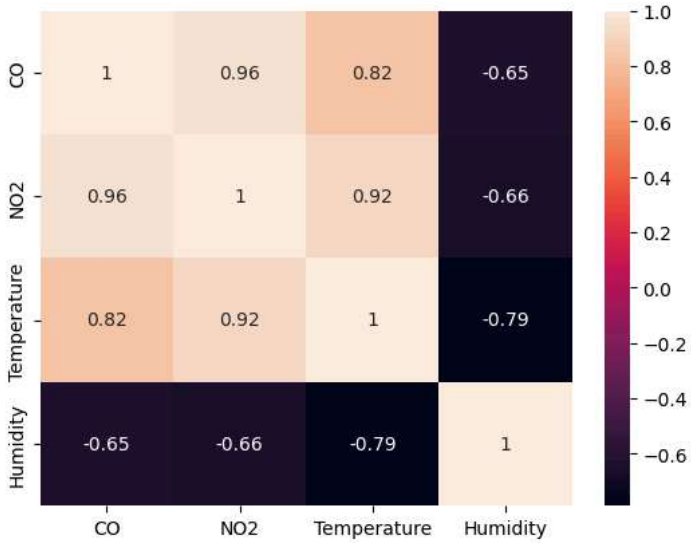
Explanation: It displays scatter plots between every pair of variables Diagonal shows distribution (histogram/KDE) Helps to: Identify correlation between variables Detect patterns and trends Find outliers

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Create sample Air Quality dataset
data = {
    "CO": [2.3, 1.8, 2.5, 3.0, 2.2],
    "NO2": [30, 25, 40, 50, 35],
    "Temperature": [25, 27, 30, 35, 28],
    "Humidity": [60, 55, 50, 45, 65]
}

df = pd.DataFrame(data)

# Pair Plot
sns.pairplot(df)
plt.show()
```



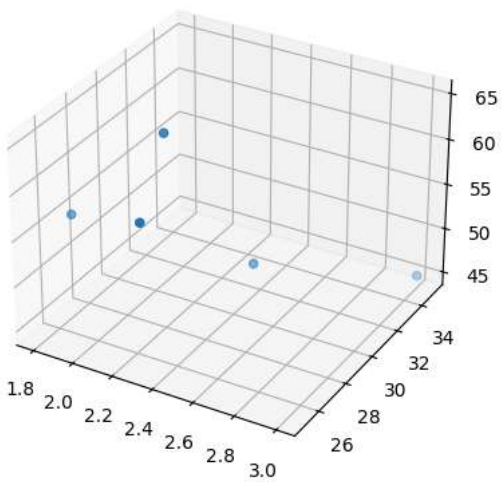
Heatmap

```
sns.heatmap(df.corr(), annot=True)
plt.show()
```



3D Scatter Plot

```
from mpl_toolkits.mplot3d import Axes3D
import matplotlib.pyplot as plt

fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

ax.scatter(df['CO'], df['Temperature'], df['Humidity'])
plt.show()
```



4 :- In Wine quality data set can you do non graphical EDA ? Explain your answer.

Ans :- Non-Graphical EDA on Wine Quality Dataset

The Wine Quality dataset contains chemical properties of wine such as:

Fixed acidity Volatile acidity Citric acid pH Alcohol Quality (output variable)

Techniques Used in Non-Graphical EDA

1. Summary Statistics Mean Median Standard deviation Minimum and Maximum

```
import pandas as pd

# Create sample Wine dataset
data = {
    "fixed_acidity": [7.4, 7.8, 7.8, 11.2, 7.4],
    "volatile_acidity": [0.70, 0.88, 0.76, 0.28, 0.70],
```

```
    "citric_acid": [0.00, 0.00, 0.04, 0.56, 0.00],
    "pH": [3.51, 3.20, 3.26, 3.16, 3.51],
    "alcohol": [9.4, 9.8, 9.8, 9.8, 9.4],
    "quality": [5, 5, 5, 6, 5]
}

df = pd.DataFrame(data)

# 1. Summary statistics
print(df.describe())

# 2. Missing values
print(df.isnull().sum())

# 3. Data types
print(df.dtypes)

# 4. Quality distribution
print(df['quality'].value_counts())

# 5. Correlation
print(df.corr())

# 6. Skewness
print(df.skew())
```

```
       fixed_acidity  volatile_acidity  citric_acid        pH   alcohol  \
count       5.000000          5.000000     5.000000  5.000000  5.000000
mean        8.320000          0.664000     0.120000  3.328000  9.640000
std         1.622344          0.226892     0.246577  0.169912  0.219089
min         7.400000          0.280000     0.000000  3.160000  9.400000
25%         7.400000          0.700000     0.000000  3.200000  9.400000
50%         7.800000          0.700000     0.000000  3.260000  9.800000
75%         7.800000          0.760000     0.040000  3.510000  9.800000
max        11.200000          0.880000     0.560000  3.510000  9.800000

        quality
count  5.000000
mean   5.200000
std    0.447214
min    5.000000
25%    5.000000
50%    5.000000
75%    5.000000
max    6.000000
fixed_acidity       0
volatile_acidity    0
citric_acid         0
pH                  0
alcohol             0
quality             0
dtype: int64
fixed_acidity       float64
volatile_acidity    float64
citric_acid         float64
pH                  float64
alcohol             float64
quality               int64
dtype: object
quality
5    4
6    1
Name: count, dtype: int64
                   fixed_acidity  volatile_acidity  citric_acid        pH  \
fixed_acidity           1.000000         -0.906283     0.994920 -0.650087
volatile_acidity       -0.906283          1.000000    -0.943763  0.281700
citric_acid             0.994920         -0.943763     1.000000 -0.577617
```

2. Missing Values Using isnull().sum() to check null values
3. Data Types Using dtypes to identify numerical features
4. Frequency Distribution Using value_counts() for quality (ordinal data)
5. Correlation Analysis Using corr() to find relationships between variables
6. Skewness Using skew() to check data distribution